OFFICIAL CORPORATE DOSSIER · 2026

ENGINEERED IN PAKISTAN
ESTABLISHED 2008

TSP TENSILE STRUCTURES (PVT) LTD

Architectural Membrane *Engineering Practice*

Pioneering Turnkey Tensile Fabric Architecture, Computational Form-Finding, Hot-Dip Galvanized Structural Steel & High-End Cantilever Shading Systems across Pakistan.

18+	150+	160 KM/H	100%
YEARS PEDIGREE	PROJECTS DELIVERED	WIND TESTED	IN-HOUSE TURNKEY

CERTIFIED ENGINEERING COMPLIANCE
ASCE 7–10 Wind Codes · DIN 4102–B1 Flame Retardant · ASTM A123 Hot-Dip Galvanizing

HEADQUARTERS: LAHORE, PAKISTAN
Islamabad · Rawalpindi · Karachi · Nationwide Operations

— PIONEERING TENSILE ARCHITECTURE

Engineering Artistry in Fabric & Steel: Structures That Belong Where They Are Placed

TSP Tensile Structures is Pakistan's premier specialized engineering practice dedicated exclusively to the architectural design, computational finite-element analysis, precision steel fabrication, and tensioned membrane installation of modern shade structures.

THE SINGLE-SOURCE TURNKEY COMMITMENT

Unlike typical metal fabricators who outsource fabric cutting or purchase substandard uncertified PVC, TSP controls the entire value chain in-house. From the first topographical laser scan and computational form-finding to robotic steel welding, 80-micron molten zinc galvanizing, and final hydraulic rigging—our unified team delivers zero handoff errors.

01. SITE FIRST

Every structure is tailored to specific sun-angles, wind dynamics, and architectural curb appeal.

02. HONEST GEOMETRY

Pure hyperbolic anticlastic curves that achieve stability through mathematical tension.

03. IN-HOUSE ONLY

Engineering, steel fabrication, and membrane RF welding under one accountable roof.

04. 25-YR LIFECYCLE

Specified materials engineered to withstand severe monsoons and brutal 48°C summer heat.



Architectural Cantilever Car Parking System — DHA Luxury Villa Installation

150+	18+ Years	99.8%	15–25 Yrs
COMPLETED INSTALLATIONS	STRUCTURAL EXPERIENCE	SOLAR UV BLOCKOUT	EUROPEAN FABRIC LIFESPAN

ENGINEERING PEDIGREE

Led by Engr. Shaukat Rauf: 18+ Years of Mega-Infrastructure Mastery

Tensile architecture is structural engineering at its most unforgiving. Under the direct leadership of Chief Executive Officer Shaukat Rauf, TSP Tensile applies rigorous international erection and fabrication standards to every tensile structure built in Pakistan.

CAREER CHRONOLOGY & PROJECT HERITAGE

HAIF TRADING & CONTRACTING — 2008 – 2014 (RIYADH, KSA)
6 years of high-load structural steel erection, industrial canopy supervision, and heavy structural assembly across major commercial and governmental infrastructure in Saudi Arabia.

SARWAR & COMPANY · KOTO HYDROELECTRIC — 2016 – 2017 (PAKISTAN)
Supervised complex structural steel erection and heavy-duty load-bearing assemblies on the landmark KOTO Hydroelectric Power Project.

FRONTIER WORKS ORGANIZATION (FWO) — 2017 – 2019 (PAKISTAN)
Erection Supervisor managing critical structural assemblies, heavy civil frames, and high-tolerance military/strategic transport infrastructure.

DESCON ENGINEERING LTD. — 2019 – 2021 (PAKISTAN)
Served as Structural Fabrication & Erection Foreman at Pakistan's largest multinational engineering enterprise, directing plant fabrication and precision welding teams.

FOUNDING CEO · TSP TENSILE STRUCTURES — 2021 – PRESENT
Spearheaded the turnkey execution of 150+ iconic tensile membrane installations spanning Radio Pakistan, PAF, KFC, Nestlé, DHA, and luxury residential estates nationwide.



In-House High-Precision CNC Steel Fabrication Yard — Lahore Facility

"In tensile architecture, compromise is lethal. A membrane structure is held together by equilibrium. If the steel is undersized, the galvanizing thin, or the fabric uncertified, nature exposes it in the very first storm. At TSP Tensile, every weld and turnbuckle is engineered with the exact same rigor I applied to mega-infrastructure in Saudi Arabia."

ENGR. SHAUKAT RAUF
Chief Executive Officer · TSP Tensile Structures

Shaukat Rauf

— ARCHITECTURAL TYPOLOGIES

Six Specialized Tensile Solutions Engineered for Form & Function

Each architectural shade system addresses distinct aerodynamic challenges, structural spans, and spatial aesthetics. We design for residential villas, institutional campuses, sports clubs, and commercial fleet facilities.

01. CANTILEVER CAR SHADES

ZERO FRONT COLUMN INTERFERENCE

Our signature product for luxury residences and corporate parking. Cantilever arms project outward up to 6.5 meters, allowing total unobstructed vehicle door swing and zero driving hazards.

Spans: 18×18 ft (2 Cars) to 100+ ft fleet rows

02. CONICAL & GAZEBO PEAKS

DRAMATIC 3D TENSION VERTEX

Classic high-peak geometry suspended from central compression rings. Ideal for estate lawns, clubhouse terraces, and rooftop gazebos requiring panoramic architectural drama.

Spans: 15×15 ft to 40×40 ft single modules

03. HYPERBOLIC (HYPAR) SAILS

SCULPTURAL ANTICLASTIC TWISTS

Opposing high and low anchoring points produce saddle-shaped surfaces that naturally shed torrential monsoon rain and resist cross-directional gale winds with minimal steel weight.

Application: Pools, Patios, Corporate Plazas

04. INVERTED UMBRELLA SHADES

INTERNAL RAINWATER DRAINAGE

A central hollow steel mast channels rainwater down an internal pipe, eliminating edge runoff. Widely specified for commercial courtyards, café patios, and institutional drop-off bays.

Spans: 16×16 ft to 25×25 ft freestanding

05. WAREHOUSE & CLEAR-SPAN

MASSIVE UNBROKEN COVERAGE

Engineered for factories, agricultural storage, and logistics facilities. Provides massive covered volume at 40% lower cost than heavy concrete, erected in a fraction of the time.

Spans: 30 ft to 120+ ft unobstructed clear spans

06. SPORTS ARENAS & PADEL

ALL-WEATHER UV ENCLOSURES

High-translucency architectural membranes allow 12% natural diffused daylight without blinding glare, keeping courts cool in summer while guaranteeing complete monsoon playability.

Application: Padel, Tennis, Basketball, Bleachers



Institutional Covered Walkway — Lahore



Estate Swimming Pool Canopy — Islamabad



Padel Court All-Weather Roof — Arena Sports

— CERTIFIED MEMBRANE TECHNOLOGY

European Architectural Fabrics: Engineered Against UV Degradation & Heat

Generic signage flex and cheap unlacquered tarpaulins disintegrate, yellow, and tear in Pakistan's brutal solar UV within 24 months. TSP exclusively imports Tier-1 certified architectural membranes from Germany and France.

GERMANY · MEHLER TEXNOLOGIES

VALMEX FR 900 GSM (TYPE II)

High-tenacity low-wick polyester base cloth with multi-layer knife coating and dual PVDF top lacquer. Repels smog, dust, and rain streaks.

- Tensile: 4300 / 4000 N/5cm
- Fire: DIN 4102-B1 Certified
- Lifespan: 15 – 20 Years
- Ideal: Cantilever Car Parking

FRANCE · SERGE FERRARI

PRÉCONTRAINT 1050 GSM (TYPE III)

Patented bi-axial tensioning during manufacturing eliminates sag, deformation, and thermal elongation under 48°C extreme heatwaves.

- Tensile: 5000 / 4800 N/5cm
- Coating: Fluotop T2 Anti-Dirt
- Lifespan: 20 – 25 Years
- Ideal: High-Span Commercial

ARCHITECTURAL PTFE FIBERGLASS

PTFE COMPOSITE 1250 GSM

Inert Teflon-coated woven fiberglass. Completely non-combustible Class A1 fire rating. Naturally self-cleaning under rain deluges.

- Tensile: 8000 / 7000 N/5cm
- Fire: Class A1 Non-Combustible
- Lifespan: 25 – 35+ Years
- Ideal: Stadiums & Civic Icons

LABORATORY TEST BENCHMARK MATRIX (CERTIFIED MILL CERTIFICATES)

Performance Parameter	Local / Generic Flex (Uncertified)	German Mehler Valmex 900	French Serge Ferrari 1050	PTFE Fiberglass 1250
Tensile Strength (Warp / Weft)	1200 / 1000 N/5cm (Weak)	**4300 / 4000 N/5cm**	**5000 / 4800 N/5cm**	**8000 / 7000 N/5cm**
Tear Resistance	150 N (Prone to tearing)	500 N (Micro-ripstop)	600 N (Bi-axial mesh)	900 N (Woven glass)
Fire Safety Rating	Highly Combustible (Toxic smoke)	DIN 4102-B1 Self-Extinguishing	DIN 4102-B1 / NFPA 701	Class A1 Non-Combustible
UV Radiation Blockout	40% – 60% (Vehicle paint fades)	99.8% Complete UV Block	99.9% Complete UV Block	100% Total Solar Shield
Surface Top Lacquer	None (Smog sticks permanently)	Dual PVDF Fluoropolymer	Fluotop T2 PVDF	100% Teflon (Fluoropolymer)
Documented Lifespan	18 – 24 Months	15 – 20 Years	20 – 25 Years	25 – 35+ Years

— CORROSION SCIENCE & STEEL ERECTION

80-Micron Hot-Dip Galvanizing: Structural Steel Engineered Against Rust

Monsoonal humidity, urban smog, and thermal expansion cause ordinary oil-painted steel shades to blister, flake, and rust within 3 years. TSP subjects all hollow structural members to certified 80–micron molten zinc submersion.

HEAVY-GAUGE ASTM HOLLOW STRUCTURAL SECTIONS

We fabricate using seamless ASTM A53 Grade B and ASTM A500 Grade B high-tensile structural pipes and rectangular hollow sections (RHS). Wall thicknesses are calculated in SAP2000 to eliminate arm deflection under 160 km/h wind shear.

01 **ASTM A123 Full-Submersion Hot-Dip Galvanizing**
Fabricated steel arms are bathed in 450°C molten zinc, creating an 80+ micron permanent metallurgical bond that self-heals surface nicks.

02 **Two-Pack Marine Epoxy & Aliphatic Polyurethane Topcoat**
Optional dual coat: Zinc-rich epoxy primer followed by an automotive-grade polyurethane topcoat matched to client RAL architectural color palette.

03 **Grade 316 Marine Stainless Steel Rigging & Turnbuckles**
All perimeter tension cables, swaged clevis studs, shackles, and turnbuckles are forged from AISI 316 marine stainless steel—zero rust staining on fabric.

04 **High-Tensile Chemical Foundation Anchoring**
Base plates (25mm–32mm solid steel) are anchored into 1:2:4 reinforced concrete footings with Hilti/Fischer chemical capsules rated for immense overturning moments.



Cantilever Frame with Galvanized Armature — Designed for 160 km/h Gale Resistance

— RIGOROUS EXECUTION PROTOCOL

The Six-Stage Turnkey Workflow: From Computational Form to Final Rigging

Every commission follows our ISO-aligned six-stage pipeline. We eliminate subcontractor handovers and deliver complete turnkey structural engineering from initial sketch to final tension certificate.

STAGE 01 · SITE MAPPING

ON-SITE 48H

Digital 3D laser measurement of boundary walls, driveways, and underground utility conduits. Sun-path shadow analysis computes exact shade coverage at peak solar azimuths.

STAGE 02 · COMPUTATIONAL FORM-FINDING

CAD / RHINO

Parametric form-finding using Rhino, Grasshopper, and Easy Tensile. Membrane equilibrium shapes are solved mathematically to ensure uniform surface pre-tension and zero water ponding.

STAGE 03 · FEA STRUCTURAL SIMULATION

SAP2000 / ANSYS

Non-linear Finite Element Analysis (FEA) per ASCE 7-10 wind standards. Structure is simulated under 160 km/h monsoonal gale gusts, live roof loads, and base anchor overturning moments.

STAGE 04 · CNC FABRICATION & GALVANIZING

IN-HOUSE YARD

Precision CNC plasma pipe cutting and certified robotic/MIG structural welding. Framework is hot-dip galvanized in 450°C molten zinc (80 microns) followed by 2-pack polyurethane coating.

STAGE 05 · HIGH-FREQUENCY RF MEMBRANE SEAMING

RADIO FREQUENCY

European membranes are CNC pattern-cut to compensating warp/weft stretch factors. Seams are molecularly fused using High-Frequency (RF) welding machines for 100% watertight joint strength.

STAGE 06 · PRECISION RIGGING & HANDOVER

10–15 YR WARRANTY

TSP certified rigging team erects structural steel, sets chemical anchors, and pretensions membranes using calibrated hydraulic jacks and turnbuckles. Formal warranty certificate issued.



32mm Solid Steel Baseplate with High-Tensile Chemical Anchor Foundation



Grade 316 Stainless Steel Perimeter Cable Rigging & Swaged Terminals

— LANDMARK PORTFOLIO

Landmark Commercial & Institutional Commissions Across Pakistan

From federal broadcasting headquarters in Islamabad to Olympic-grade sports complexes in Lahore, our structures serve high-profile corporate, educational, and governmental institutions.



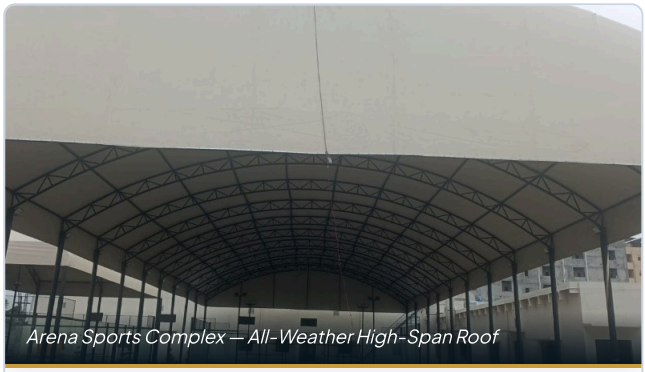
Radio Pakistan HQ — Islamabad Federal Territory

RADIO PAKISTAN HEADQUARTERS

FEDERAL BROADCASTING COMPLEX · ISLAMABAD

Engineered multi-bay heavy cantilever parking canopies sheltering executive staff and broadcast fleet vehicles. Designed to withstand severe Potohar plateau hail and monsoonal wind gusts.

Specification: German Mehler Valmex 900 GSM · 80-Micron Hot-Dip Galvanizing · 0% Column Interference


Arena Sports Complex — All-Weather High-Span Roof

ARENA SPORTS COMPLEX & PADEL CLUB

12,000 SQ.FT ALL-WEATHER CANOPY · LAHORE

Full-span architectural enclosure covering championship padel and multi-sport courts. Engineered for 12% translucent daylighting, acoustic reverberation control, and full weatherproofing.

Specification: French Serge Ferrari Précontraint 1050 GSM · Heavy Trusses · 160 km/h Wind Safety

NOTABLE INSTITUTIONAL & CORPORATE COMMISSIONS

- Pakistan Air Force (PAF) Bases
- Government House Murree
- Nestlé Pakistan Facilities
- KFC Drive-Thru Canopies
- Emporium Tower Lahore
- Hamid Latif Hospital
- Beaconhouse National University
- ECS Flagship Corporate HQs

— LUXURY LIVING & FUTURE MOBILITY

Residential Villas, Luxury Estates & High-Capacity EV Infrastructure

Protecting multi-million rupee luxury vehicles and architectural outdoor spaces across Pakistan's most prestigious residential enclaves, including DHA Lahore, Bahria Town, and Islamabad sectors.



DHA Phase 5 & 6 Luxury Villa Cantilevers — Lahore

DHA LUXURY RESIDENTIAL VILLAS

PHASE 5 & 6 ARCHITECTURAL CANOPIES · LAHORE

Bespoke 2-car and 3-car cantilever driveway shades designed to complement modern desert-stone architecture. Preserves automotive leather interiors and paint from blistering 45°C summer sun.

Specification: Serge Ferrari 1050 GSM · Champagne Metallic · Full 80-Micron Molten Galvanized Steel



Commercial EV Charging Superhub — High-Span Canopy

COMMERCIAL EV FAST CHARGING SUPERHUB

HIGH-CLEARANCE MULTI-BAY CHARGER COVER

High-clearance tensile shade structure designed for EV rapid chargers and customer waiting bays. Incorporates concealed rainwater downspouts and hidden electrical LED conduits.

Specification: German Mehler Valmex 900 GSM · Solar-Compatible Framework · Heavy Base Anchorages

APPROVED INSTALLER IN PREMIER HOUSING SOCIETIES

- DHA Lahore (Phases 1-9)
- Bahria Town (Lahore & RWP)
- Gulberg & Model Town
- Islamabad Sectors F-6, F-7, F-8
- DHA Islamabad & Bahria Enclave
- DHA Karachi & Clifton
- Lake City & Valencia Lahore
- Cantt & Officers Housing Schemes

— TRANSPARENT TURNKEY INVESTMENT

Commercial Pricing Matrix (Turnkey Rates in PKR Per Sq.Ft)

We maintain absolute transparency. All quoted prices include complete 3D architectural engineering, heavy structural steel fabrication, certified hot-dip galvanizing, imported European membrane, and on-site certified erection.

Structure Typology	Recommended Membrane	Turnkey Rate (PKR / Sq.Ft)	Design Life	Typical 2-Car Package
Standard Cantilever Car Shade	German Mehler Valmex 900 GSM	**PKR 450 – 520 / sq.ft**	15–20 Years	**PKR 160,000 – 185,000**
Heavy Architectural Cantilever	French Serge Ferrari 1050 GSM	**PKR 580 – 680 / sq.ft**	20–25 Years	**PKR 195,000 – 230,000**
Conical & Gazebo Peak Pavilion	Mehler Valmex FR 900 / Ferrari	**PKR 550 – 750 / sq.ft**	15–20 Years	**Custom Estate Quote**
Inverted Umbrella (Center Mast)	Serge Ferrari Précontraint 1050	**PKR 650 – 850 / sq.ft**	20–25 Years	**PKR 220,000 – 280,000**
Swimming Pool & Terrace Hypar	Dual Tone European PVDF	**PKR 500 – 700 / sq.ft**	15–20 Years	**Custom Geometric Quote**
Institutional PTFE Fiberglass Roof	Architectural PTFE 1250 GSM	**PKR 1,200 – 1,800 / sq.ft**	25–35+ Years	**Civic & Stadium Tender**

WHAT IS 100% INCLUDED IN OUR RATES:

- Digital topographical site survey and sun-path mapping
- 3D computational form-finding & visual render
- SAP2000 structural wind load calculation report
- Heavy ASTM steel pipe armature & 32mm base plates
- 80-micron Hot-Dip Galvanizing (molten zinc bath)
- Two-pack epoxy primer + polyurethane topcoat
- Original imported European membrane with mill certificates
- Complete rigging, turnbuckles & chemical anchor bolts

LIVE ESTIMATION CALCULATOR ON WEBSITE:

Clients and architects can calculate precise custom quotations in real-time on our interactive online cost estimator. Select your city (Lahore, Islamabad, Karachi), fabric weight, and structure typology for immediate PDF breakdown.

Visit: **www.tsptensile.com/calculator**

— GUARANTEED ENGINEERING COMPLIANCE

International Engineering Codes & Written 10–15 Year Corporate Warranty

Every tensile structure constructed by TSP complies with recognized international building codes and fire-safety criteria. We stand behind our craftsmanship with an official corporate warranty seal.

ASCE 7-10 & BCP 2021

160 KM/H WIND CERTIFICATION

Engineered per Building Code of Pakistan (BCP 2021) and American Society of Civil Engineers standards to withstand severe monsoonal storms, wind uplift, and heavy hail impact.

DIN 4102-B1 & NFPA 701

FLAME RETARDANCY

All imported European membranes are chemically treated to be self-extinguishing. In the event of open flame contact, fabric will not drip molten plastic or propagate combustion.

ASTM A123 & ISO 1461

HOT-DIP GALVANIZING

Standard specification for zinc (hot-dip galvanized) coatings on structural steel. Guarantees 80 microns minimum metallurgical thickness for multi-decade rust immunity.

THE TSP TENSILE WRITTEN WARRANTY GUARANTEE

Every client receives an official serialized Warranty Certificate signed by the Chief Executive Officer. Our guarantee covers fabric integrity against UV degradation, seam delamination, steel weld fractures, and structural failure under specified wind parameters for a period of up to 15 years.

- ✓ Free 1st Year Annual Tension Inspection
- ✓ 24-Hour Emergency Storm Response
- ✓ Registered Corporate Warranty Seal



NATIONWIDE DEPLOYMENT & CONSULTATION

Let Us Engineer Your Vision
Book an Architectural Survey

LAHORE HEADQUARTERS & FABRICATION YARD

Multan Road / Raiwind Road Industrial Area, Lahore, Punjab.
Direct Survey Desk: 0302 4001063
Email: info@tsptensile.pk

ISLAMABAD & RAWALPINDI REGIONAL DESK

G-11 Markaz / Bahria Town Phase 7, Rawalpindi / Islamabad.
Site Survey Lead: 0302 4001063
Covering: Islamabad, Rawalpindi, Peshawar, Abbottabad

KARACHI COASTAL DIVISION

Clifton & DHA Karachi Project Office.
Specialized marine-grade anti-saline tensile engineering.
Contact: 0302 4001063

DIRECT EXECUTIVE WHATSAPP DESK

Engr. Shaukat Rauf (Chief Executive Officer)
WhatsApp Hotline: +92 302 4001063
Web: www.tsptensile.com

TSP TENSILE STRUCTURES (PVT) LTD

Pakistan's #1 Turnkey Tensile Architectural Membrane Manufacturer

© 2026 TSP Tensile Structures. All Rights Reserved.
Designed by Senior Architectural Brand Practice.